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Data governance in metaverse: Addressing security threats and countermeasures across the data lifecycle

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ABSTRACT

-In the era of the Fifth Industrial Revolution, metaverse embodies a sophisticated fusion of advanced technologies, introducing a multitude of complexities and distinctive vulnerabilities. This paper undertakes a systematic review of 76 scholarly articles to explore the critical data security issues associated with the metaverse. By analyzing each phase of the data lifecycle—encompassing acquisition, storage, transfer, processing, exchange, and destruction—the study identifies key threats and evaluates corresponding mitigation strategies. A notable contribution of this research is the comprehensive categorization of threats across all lifecycle phases, highlighting vulnerabilities stemming from the cutting-edge technologies underlying the metaverse. Additionally, this study emphasizes the importance of exploring solutions within the realm of infrastructure technologies. By emphasizing data governance and safeguarding data flow throughout its lifecycle, this study significantly contributes to the ongoing discourse on metaverse security, fostering the development of a more resilient and trustworthy virtual ecosystem.

1. Introduction

Metaverse, as an emerging concept for technology interpreters, fundamentally diverges from encapsulating the internet or any existing social media platforms. This evolving concept has transformed into an immersive three-dimensional (3D) online social platform, introducing innovative user experiences (Hollensen et al., 2022). Metaverse can be defined as democratic, decentralized, user-centric, and enhanced 3D virtual spaces facilitating a more tangible connection between two worlds – the virtual and the physical. This enables individuals, regardless of their physical locations, to converge and interact (Dwivedi et al., 2022). The metaverse is a "combination of virtual and physical entities" aimed at developing virtual worlds to enrich the real world with high fidelity (Kuru, 2023). To integrate the virtual and real worlds in the metaverse, creating an open and collaborative chain for transferring resources between multiple virtual worlds and seamless connectivity through Web 3.0 is crucial. Physical interactions between real and reconstructed 3D objects can be facilitated through active sensors. Additionally, passive reconstruction may involve no interactive elements between objects (Jagatheesaperumal et al., 2023).

To comprehend the physical world from the perspective of metaverse platforms, Artificial Intelligence (AI) aids in orchestrating workflows and participating in collaborative operations by enhancing and automating tasks in virtual spaces. Metaverse platforms, equipped with Internet of Things (IoT) and Digital Twins (DTs), facilitate movement in the virtual world and gather relevant information about their digital peers based on demand. Support and interaction with DTs and immersive capabilities are made possible through Extended Reality (XR) solutions that realize an integration of the physical and virtual worlds. Achieving this integration requires significant computational power, the deployment of intelligent edge devices, and robust network infrastructure (Lam et al., 2022). Blockchain and cryptocurrencies act as the economic backbone of the metaverse, bridging real-world and metaverse interactions, and facilitating seamless interactions within the virtual environment (Trong et al., 2023).

The convergence of these technologies (IoT, DTs, AI, XR, and blockchain), while significantly contributing to the realization of the metaverse, poses a substantial challenge. The metaverse is likely to inherit security and privacy breaches stemming from these technologies. The metaverse faces a diverse range of security vulnerabilities stemming from the handling of extensive data flows, widespread user profile activities, biased outcomes of AI algorithms, and concerns regarding the safety of both physical infrastructure and human bodies. Converging these technologies brings new emerging risks such as potential hijacking

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of wearable devices or cloud storage spaces, cryptocurrency theft, and the malicious utilization of AI to generate fake news.

The metaverse constitutes a substantial source of data, consistently growing in both quantity and variety. Data from various sources such as users, devices, and sensors are transferred to DTs, employing either processing methods or direct transmission. The process of transferring data from the metaverse to the physical world also takes place. Thus, the continuous exchange of data and feedback between the physical world and the metaverse fosters their convergence, seamlessly uniting these two interconnected realms. Recognizing that the metaverse's vulnerabilities and targeted attacks are intricately connected to its data and their flow, this article aims to explore these vulnerabilities from the standpoint of effective data governance.

Data governance encompasses a set of policies, processes, and controls that each organization/application is tasked with to protect data and ensure security from a triadic perspective (confidentiality, integrity, and availability). Managing secure data throughout its lifecycle involves multiple stages that organizations interact with regarding data governance. These stages are approached differently by various frameworks (Cheng, 2023), (Soliman et al., 2023), (Pouchard, 2015). For instance, IBM's data lifecycle model encompasses data creation, use, analysis, sharing, updating, protection, archiving, storage, and disposal (IBM & Wrangling Big Data, 2013). Similarly, the big data lifecycle model, as outlined by the National Institute of Standards and Technology (NIST), includes phases like collection, preparation, analysis, and action (analytics, visualization, access) (Group, 2015). In industrial and healthcare applications, the data lifecycle typically involves creation, storage, analysis, usage, sharing, archiving, and destruction (Rahul & Banyal, 2020). Meanwhile, for IoT applications, the data lifecycle is often categorized into phases like data acquisition, processing, storage, utilization, and dissemination (Soltani Panah et al., 2019).

To investigate and identify vulnerabilities within the data lifecycle in the metaverse, this study adopts the secure data governance framework (Hussain Shah et al., 2021) as a guiding reference. Therefore, to explore and identify vulnerabilities within the data lifecycle in the metaverse, we employ the secure data governance framework as our guiding reference. In the metaverse, the data lifecycle encompasses six stages: acquisition, storage, transfer, processing, exchange, and destruction. This article examines the vulnerabilities and security threats present at each of these stages. Additionally, strategies for mitigating and addressing these risks are thoroughly analyzed and discussed. To achieve a comprehensive and detailed analysis of all identified vulnerabilities and their corresponding mitigation measures, a systematic review approach has been employed. The primary objectives of this research are.

- To identify the security threats associated with data at each stage of its lifecycle.
- To determine effective countermeasures for addressing these threats at every stage of the data lifecycle.

The article is structured as follows: In Section 2, we examine previous related works in the domain of data security within the metaverse. Section 3 introduces the research methodology. In Section 4, we delve into identifying security attacks at various stages of the data lifecycle. Strategies for mitigating these attacks at each stage are elaborated in Section 5. Furthermore, we discuss ongoing challenges in data security in Section 6, and finally, Section 7 presents the conclusion of the article.

2. Related works

As the concept of metaverse gains prominence, there has been a consistent effort to delve into the metaverse and analyze its various vulnerabilities. Addressing and resolving these vulnerabilities can foster greater trust and significantly enhance the user experience (Al-kfairy et al., 2023). In the majority of research conducted in 2022, there has been a combined focus on data security and user privacy. By examining information flows (Canbay et al., 2022), has investigated risks associated with privacy. Investigating industrial advancements in the metaverse realm (Choi et al., 2022), has delved into data security and privacy concerns. With a focus on various applications of the metaverse (Ooi et al., 2022), has examined the diverse managerial and engineering challenges in the metaverse data, including data fusion, data distribution, database engines, data consistency, decision-making, and data streaming.

Most review studies published in 2023 focus on analyzing and classifying the vulnerabilities of the metaverse, exploring its various dimensions from a comprehensive perspective. For example (Wang et al., 2023), has classified these vulnerabilities into seven primary categories including authentication and access control, data management, privacy-related concerns, network-related issues, economy-related challenges, physical/social effects, and government-related aspects. Then, in each category, threats and security countermeasures have been explored. By examining the metaverse within an organization context (Li, Yao, et al., 2023), has conducted on security threats and proposed some solutions for ensuring robust security measures (Cheng, 2023). has focused on analyzing some security threats in the metaverse infrastructure. The approach presented in (Dunnett et al., 2023) has extensively observed security threats associated with the metaverse. Concentrating on the datasets associated with metaverse users, the study has scrutinized security threats related to their privacy. Focusing on data (Singh, 2023), has examined security threats in a digital environment.

In contrast, most review studies published in 2024 focus on the individual components of the metaverse, seeking to understand their impact on its specific security issues. For example (Awadallah et al., 2024), explores AI-driven cybersecurity techniques for the metaverse, examining academic and industrial perspectives. It focuses on three critical aspects including user authentication, intrusion detection, and the security of digital assets, specifically for blockchain-based metaverse and Non-fungible Tokens (NFTs). By acknowledging XR as a key infrastructure of the metaverse (Kourtesis, 2024), investigates the associated risks, vulnerabilities, and challenges. Beyond technical solutions like AI and regulations to address these issues, the study also explores strategies to foster creativity, promote mental independence, and ensure equitable access to technology (Sharma et al., 2024). explores the threats and vulnerabilities that metaverse users may encounter, identifying key security gaps. In response, it proposes a new architecture built on zero-trust principles and blockchain technology, with a focus on prioritizing user control over their data, identity, and experiences (Eltanbouly et al.). focuses on identifying the different threats that avatars may encounter in the metaverse. These threats are categorized into two main types: identity-related risks, such as identity theft, and social threats, such as harassment. To mitigate these risks, the study outlines a set of strategies, organized into three areas: modifying user representations, providing safety options for users and avatars, and utilizing AI to detect and address potential issues. Through a case study examining the metaverse in a market context (Zhang et al., 2024), has focused on data management, so it explored challenges and discussed some methods for ensuring data security.

Table 1 provides a summary of previous contributions.

In this article, a comprehensive examination of the metaverse is presented, encompassing its data security threats, and solutions to counter them. The contributions of this article are summarized in the following areas.

- This study explores security data threats at each phase by analyzing different stages in the metaverse data lifecycle.
- By introducing security countermeasures, this paper delineates prospective research directions for establishing a secure and efficient metaverse environment.

Table 1
A summary of contributions from relevant surveys.

Year	Reference	Contribution
2024	Awadallah et al. (2024)	Survey on AI-based cybersecurity solutions for the metaverse, exploring user authentication, intrusion detection, and asset trading.
2024	Kourtesis (2024)	This paper reviews and categorizes the various risks associated with XR technology and proposes solutions to address them.
2024	Sharma et al. (2024)	It provides a comprehensive study of the threats that impact user privacy and categorizes protective solutions into eight distinct groups.
2024	(Eltanbouly et al.)	Focusing on avatars, the study identifies two categories of related threats and proposes solutions to mitigate them.
2024	Zhang et al. (2024)	It examines the data management potential and challenges, including storage, indexing, retrieval, and query processing in various metaverse applications.
2023	Wang et al. (2023)	A comprehensive analysis has been presented regarding security threats categorized into seven groups.
2023	Li, Yao, et al. (2023)	This article discusses the metaverse infrastructure and potential security threats associated with it.
2023	Cheng (2023)	It explores several significant security threats.
2023	Dunnett et al. (2023)	It explores the advantages of sharing information on prevalent cybersecurity threats within the metaverse interactions.
2023	Singh (2023)	It examines the key challenges and risks associated with securing data in the new digital environment.
2022	Canbay et al. (2022)	It has examined threats related to privacy.
2022	Choi et al. (2022)	It explores security and privacy issues in Virtual Reality (VR) and Augmented reality (AR) environments.
2022	Ooi et al. (2022)	By introducing the metaverse applications, it explores data management challenges.

3. Research methodology

The term "metaverse" was first introduced in Neal Stephenson's 1992 novel "Snow Crash." However, it didn't gain significant attention until 2021, when Facebook rebranded itself as Meta. Despite prior research into key components like AI, DTs, IoT, and XR, it was the convergence of these technologies under the banner of the metaverse that truly sparked a new era of exploration. As a result, current research initiatives prioritize articles published and reviewed post-2021.

In order to investigate security challenges throughout the lifespan of data in the metaverse environment, we conducted a systematic review using the PRISMA method. To identify keywords surrounding this topic we examine this domain by reviewing security literature. Our study is confined to English-language articles, thus the inclusion criteria of the articles under scrutiny in this research are delineated below.

- Publication year: 2022 to 2024
- Language of the article: English
- Main keyword: Metaverse
- Critical keywords investigated to identify challenges: Secure or security or cybersecurity or vulnerability or threat or risk or defense or challenge or attack

While the primary focus of current research is on the security of the data lifecycle in the metaverse, the chosen keywords for article search extend beyond data security. This is due to two reasons.

1. Some articles exploring security issues in the metaverse also touch upon data security concerns.
2. Specific vulnerabilities or threats identified in the metaverse, originating from data vulnerabilities, are not explicitly labeled under data security or threats.

Hence, by expanding the research scope, our objective was to include and examine maximum instances of data security within our study wherever possible. Additionally, while our primary goal is to identify security threats, we aim to discover countermeasures. We have not explicitly targeted countermeasures in our search keywords because articles discussing security measures will naturally include terms like "security," "attack," or "threat".

Despite many reputable academic repositories available, we chose Google Scholar as the starting point for our research for several key reasons.

1. Research on the metaverse and its security issues is still emerging. Thus, we may find preprint articles on arXiv that haven't been published yet.
2. We aim to avoid restricting ourselves to specific types of articles. Relevant information for our research could be found in conference papers, journal articles, book chapters, reputable reports, and even lecture notes.
3. The metaverse encompasses a wide range of technologies, including AI, IoT, blockchain, DTs, and XR. Given this broad scope, limiting our search to a few repositories could hinder the comprehensiveness of our investigation.

Therefore, we selected Google Scholar, which indexes articles from all fields and various types of publications, ensuring an extensive search. In response to the entered query, a total of 367 articles were retrieved. Pursuing the defined objective of this research, the abstracts of these articles were scrutinized, leading to the selection of 75 articles as reference materials for further study. Many articles eliminated at this stage were focused solely on attacks affecting personal or system security (articles centered around authentication, access control, and user privacy) or the social security of metaverse users. Legal challenges also garnered significant attention in the retrieved articles, which were subsequently excluded from further studies. Fig. 1 provides an overview of the retrieved articles on this matter, showcasing a categorization of diverse security concerns documented within them. This article exclusively focuses on analyzing concerns within the domain of data governance.

Considering the emerging nature of the metaverse domain, numerous articles have aimed to define its concept and potential challenges following its evolved form. Consequently, there was considerable overlap in the content presented by these articles. Therefore, the list of reference articles was organized based on their citation counts for thorough examination. During the review process, articles that did not present new challenges were excluded from the reference list. Ultimately, 37 articles were identified that presented various challenges throughout the data lifecycle in the metaverse.

While examining reference articles, their associated citations were also scrutinized to identify challenges in the metaverse attributed to inherited infrastructure technology challenges or the combination of such technologies. With the inclusion of relevant citations addressing

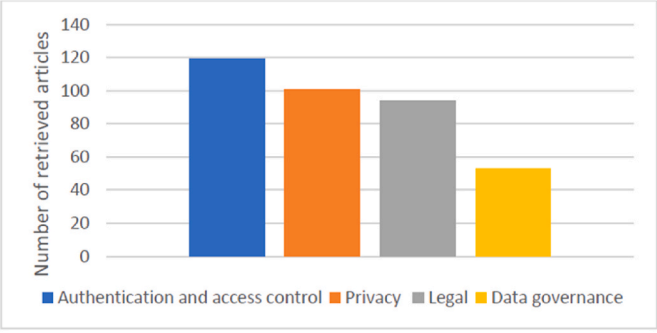


Fig. 1. Frequency of research conducted on metaverse security and challenges.

other vulnerabilities or countermeasures, the collection of reviewed articles expanded to 76. The attacks and vulnerabilities highlighted in each reviewed article were classified according to the specific stage of the data lifecycle in the metaverse where they take place. Ultimately, by analyzing these studies, the probable attacks documented in reputable sources up to the time of this writing were identified and mapped to the six stages of the data lifecycle. A similar approach was applied to countermeasures, which were systematically categorized based on their corresponding attacks and the specific lifecycle stage where those attacks take place.

To complete the reliability and validity of the included studies, each article was evaluated based on its methodological rigor, potential conflicts of interest, and clarity of findings. The credibility of the article's publisher and the authors' affiliation with universities or research institutions serve as key indicators for assessing the absence of conflicts of interest. To evaluate methodological rigor and the clarity of findings, articles were reviewed based on the following criteria. Articles meeting at least one of these criteria were included in the study.

- Identified vulnerabilities must be supported by evidence or scientific reasoning, with a detailed and convincing explanation of the observed gap or proposed countermeasure.
- The introduced vulnerability or threat must have been recognized through practical evaluation and testing (this also applies to countermeasures).
- A theoretical proof must be provided for the identified vulnerability, or a prevention strategy must be proposed.

The first approach is more commonly observed in the articles analyzing the future metaverse, which has not yet been fully realized. These studies evaluate potential vulnerabilities by synthesizing various pieces of evidence. The second approach is primarily seen in articles that focus on practical experiments or simulations, often targeting specific metaverse infrastructures. The third approach is typically employed in theoretical analyses, which, in many cases, can be adapted into actionable, practical solutions.

Fig. 2 illustrates the research process of this article.

As illustrated in Fig. 2, a total of 37 articles, which are specifically focused on the metaverse, were selected as the core foundation for our study. The remaining articles were identified through the reference lists of these core papers and predominantly address the underlying technologies of the metaverse. Fig. 3 provides an analysis of the distribution of the reviewed articles by publication year.

The articles examined in this paper, which focus on vulnerabilities in the metaverse from a data perspective, primarily span the period from 2022 to 2024. As illustrated by the trend in Fig. 3, research on vulnerabilities reached its peak in 2023. However, by 2024, the emphasis has shifted away from the metaverse concept itself, with increasing attention directed toward addressing vulnerabilities in the underlying

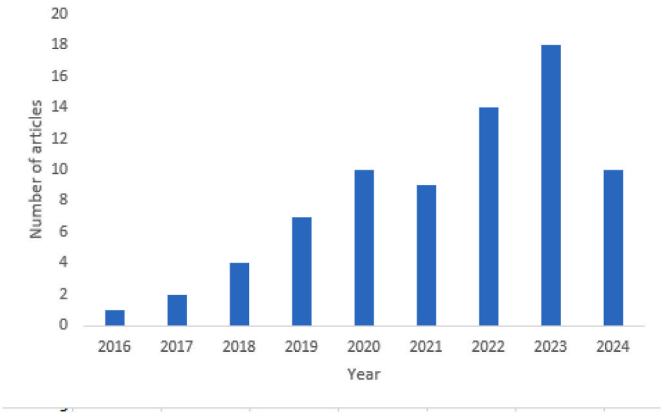


Fig. 3. The distribution of the reviewed articles based on publication year.

infrastructure technologies. This shift is further corroborated by the findings of this study, as well as additional searches conducted online. Among the reviewed articles, 53 were journal papers, 19 were conference papers, and 4 were from other sources, including reports and books.

4. Existing threats from the perspective of the secure data lifecycle

Taking a comprehensive view of the data lifecycle across various applications and research studies (Hussain Shah et al., 2021), has proposed a framework for a secure data lifecycle. This framework comprises ten stages, with five stages - collection, preparation, analysis, sharing, and usage - being mandatory, and five other stages named planning, visualization, access, archive, and termination being optional. Aligning this framework with the metaverse and its data traversal stages, the metaverse's data lifecycle includes acquisition (where both collection and preparation stages occur together at endpoints), storage, sharing or transfer, process, usage which occurs in data exchange, and ultimately, termination or destruction.

In its capacity as either an organization (Nateghi & Mosharraf, 2023) or an application framework, the metaverse manages an extensive collection of data. Consequently, it is crucial to investigate the secure data lifecycle within it, mitigating security threats at each stage. The subsequent exploration focuses on security threats at each phase of the data lifecycle in the metaverse. Fig. 4 provides a concise overview of the identified threats at each stage.

4.1. Data acquisition

The data acquisition terminal in the metaverse includes not only AR/VR devices but also a multitude of IoT devices. From a security

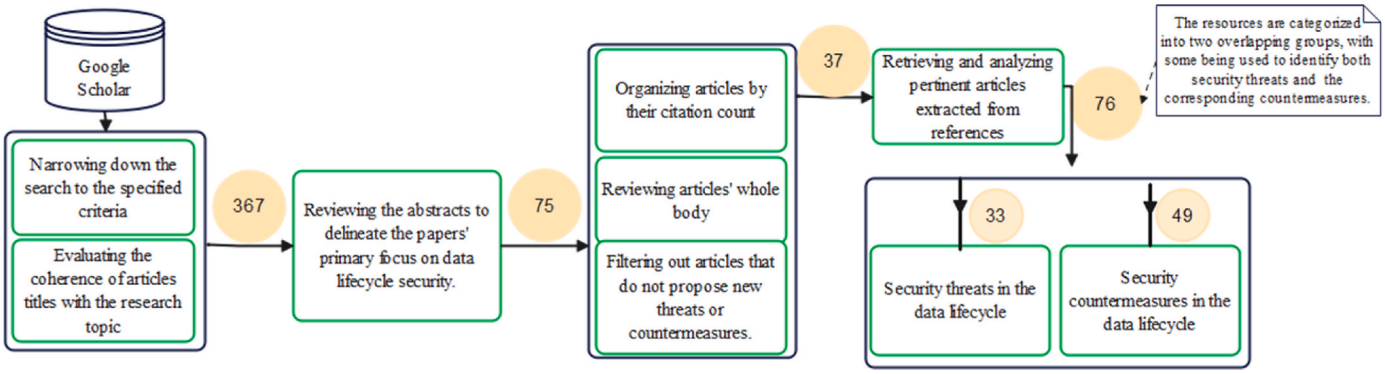


Fig. 2. The flow diagram for the paper's systematic review.

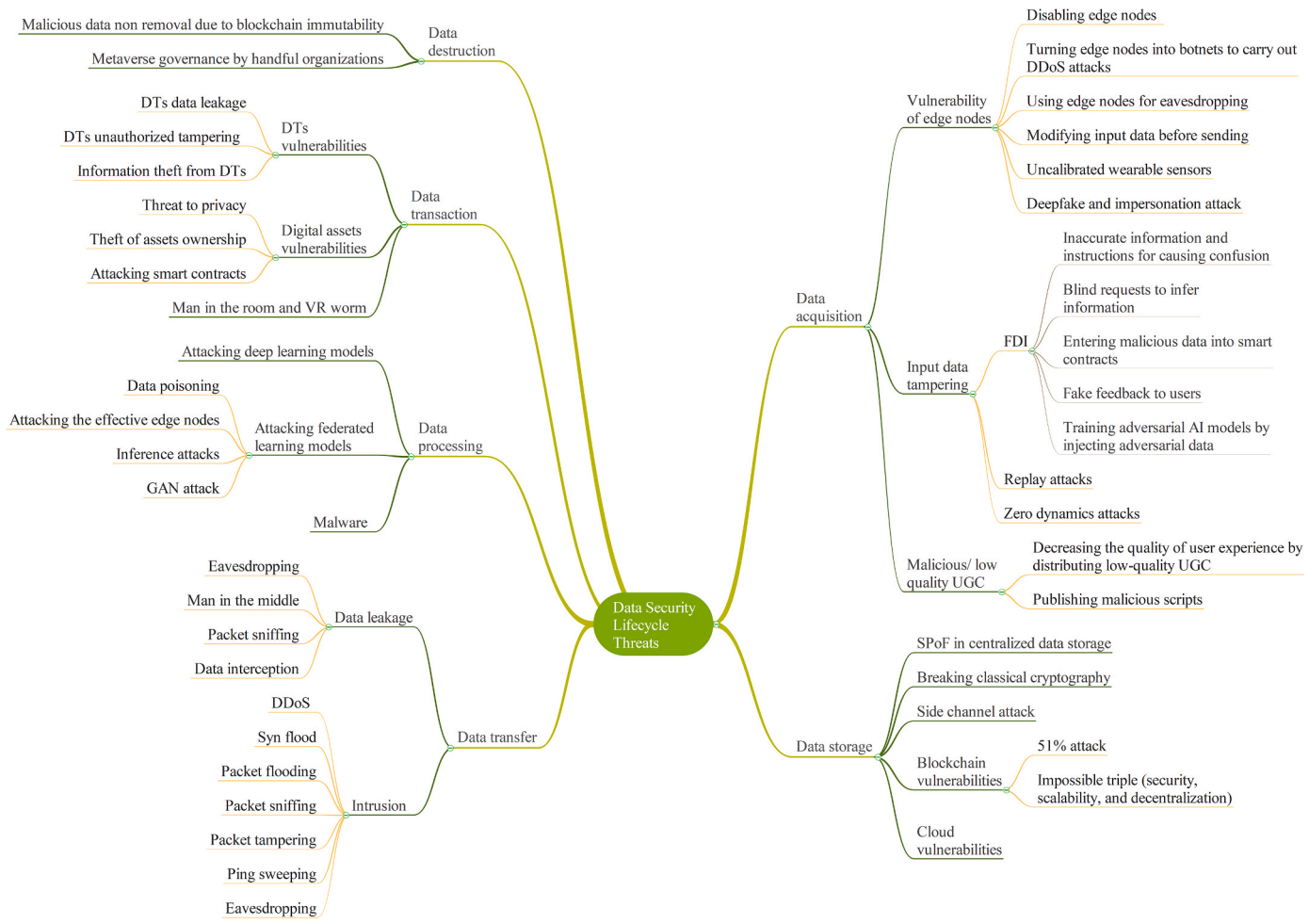


Fig. 4. Security threats in the life cycle of the metaverse data.

perspective, an increased number of endpoints exposes the system to higher levels of potential attacks (Li, Yao, et al., 2023). Data acquisition terminals, characterized by their substantial numbers, diverse types, and numerous data, are vulnerable to hacker attacks during the data gathering process. Metaverse entry points, serving both users and objects, should act as locations for managing data acquisition security.

Vulnerability of edge nodes: Many AR/VR and IoT devices lack security considerations from their initial designs. These devices, due to their portability and weak security controls, are prone to becoming targeted vulnerabilities that attackers may exploit for network infiltration (Li, Yao, et al., 2023). For instance, an attacker could leverage a network of IoT devices to autonomously distribute malware that can quickly proliferate within the metaverse. Attacks that can occur on edge nodes in the metaverse include.

- Multiple attacks on edge nodes lead to their incapacitation and disruption of data entry (Wang et al., 2023)
- Seizing control of edge nodes and turning them into botnets to conduct Distributed Denial-of-Service (DDoS) attacks (Bertino & Islam, 2017)
- Utilizing edge nodes for eavesdropping (Shang et al., 2022)
- Altering data gathered by nodes before transmission (Canbay et al., 2022)
- Creating inaccurate sensory data through uncalibrated wearable sensors and deceiving DTs (Wang et al., 2023)

Deepfake and impersonation attack is another form of attack stemming from vulnerabilities in edge nodes. To achieve this, attackers utilize AI technology to create synthetic human-like features, substituting them for authentic data and engaging in fraudulent activities (Qamar et al., 2023). In this attack, the perpetrator leverages identity forgery to gain access, manipulate information, and exploit the legitimate user's rights.

Input data tampering: Data tempering or data poisoning is a form of attack that impacts the computational power of the central controller (Liang et al., 2017) and is primarily divided into three categories, including False Data Injection (FDI) attacks, replay attacks, and zero-dynamic queue attacks (Cui et al., 2020). In FDI attacks, an attacker can inject counterfeit data into the system as much as possible without immediate detection, making real-time identification challenging. In replay attacks, the attacker repeatedly uploads hidden data to terminal devices within a specified time. By eavesdropping on data, the attacker deceives authorities by repeatedly sending the same data. Zero-dynamic queue attacks refer to exploiting unstable zeros as a vulnerability to attack the system.

Attackers can introduce false information, such as misleading messages and inaccurate instructions, into the metaverse edges. For instance, using AI to generate content may aid users in the initial phases of the metaverse. At the same time, adversaries might inject adversarial examples or manipulated gradients during both centralized and distributed AI training to produce malicious AI models. Additionally, deceptive feedback or recursively incorrect instructions can pose threats to the integrity of physical equipment and even personal safety. For example, counterfeit feedback, like an excessive output voltage, can result in damage and disruption to the functionality of wearable XR devices (Wang et al., 2023). Another scenario involves manipulated

exaggeration of pain during in-game shooting, potentially leading to real-life harm or fatality for a human user. FDI serves various purposes, including.

- Inaccurate information and instructions for confusion (Cui et al., 2020)
- Sending blind requests to infer information (Cui et al., 2020)
- Injecting malicious data into smart contracts of the metaverse infrastructure
- Sending fake feedback to users (Wang et al., 2023)
- [32] Training adversarial AI models by injecting adversarial data

Malicious/low-quality User Generated Content (UGC): One-way data enters the metaverse is when users or avatars generate content and input it directly into the metaverse environment. Selfish users/avatars may create and share low-quality content to save on their expenses, consequently diminishing the authentic experience of users in a virtual environment. Another threat stemming from UGC hosted in the metaverse is the presence of malicious scripts. Malicious users can generate harmful content or scripts and sell them in the metaverse or share them freely with other users. The dissemination of such scripts can lead to fraud or misuse of the metaverse environment and also degrade the quality of user experiences (Wang et al., 2023). Multiple attacks on platforms like Unity, Roblox, and Unreal Engine have been reported in this domain (Qamar et al., 2023).

4.2. Data storage

The increasing data volume within the rapidly expanding metaverse demands elevated standards for storage capacity, swift read and write speeds, and resilient data backup capabilities. Real-time interaction among users and avatars requires accelerated performance and heightened security. In addition, the diversity and complexity of UGC and their intricate relationships demand a shift from a single-file model to various semi-structured or unstructured data models. Despite the associated high infrastructure costs, these features make it challenging for centralized storage methods to adapt to the volume and diverse nature of the metaverse data. Additionally, the risks of data leakage, manipulation, or loss are significantly elevated. Consequently, centralized storage proves impractical for the metaverse (Chen et al., 2024). Centralized storage introduces the risk of a critical Single Point of Failure (SPoF) and increases the potential for powerful entities to exert control (Wang et al., 2023).

Secure distributed data storage in the metaverse leverages two key advantages. Firstly, advanced cryptography methods are utilized to encrypt sensitive data, serving as a deterrent against direct theft attempts by hackers. Secondly, the expansion of distributed storage capacity, coupled with enhancements in backup and recovery mechanisms, proves instrumental in ensuring the adequate availability and retrieval of data within the metaverse. Nonetheless, challenges persist in securing metaverse data storage, including:

Breaking classical cryptography: The future emphasis on storage in the metaverse has shifted towards blockchain technology (Nateghi & Mosharraf, 2023). Due to the rapid development of quantum computing technology, classical encryption systems employed in blockchains have become less secure.

Side-channel attack: Another potential threat to the blockchain infrastructure of the metaverse is a side-channel attack. In this hardware-focused attack, information is not directly leaked but is indirectly stolen from the storage location (Ghandali et al., 2021). For instance, various methods have been employed to detect the private key, such as voltage manipulation or electromagnetic wave analysis.

Blockchain vulnerabilities: Blockchains manage personal digital assets. The performance of blockchains is relatively low due to the "impossible triangle" concept, which balances security, decentralization, and scalability. It has faced criticism from users regarding its ability to ensure security and scalability. The increase in the number of Decentralized Applications (DApps) on blockchains and the growing complexity of these applications often result in network congestion and, in some cases, the unavailability of blockchain services (Khan et al., 2021). Furthermore, despite the use of common consensus mechanisms such as Practical Byzantine Fault Tolerance (PBFT), Proof of Work (PoW), Proof of Stake (PoS), and Delegated Proof of Stake (DPoS) for proposing blocks and recording data in blockchains, there are inherent security risks associated with these algorithms. For instance, PoW may be vulnerable to a double spending attack, where an attacker surpasses 51 % of the total computational power of the blockchain, granting them control over the entire blockchain (Chen et al., 2022).

Cloud vulnerabilities: Users can store some of their data in the cloud and share it with others. Authorized users can access this data at any time and from any location. While cloud infrastructures are susceptible to large-scale traffic attacks like DDoS, unauthorized access to stored data in the cloud, data leaks, misconfigurations of cloud service environments, and insufficient security are prevalent threats in these environments (Cheng, 2023).

4.3. Data transfer

The metaverse serves as an expansive network, providing users with entry into the virtual domain through VR devices. Users connect with each other or their avatars, participate in virtual commerce, undergo medical consultations via telemedicine, join virtual classes and laboratories, and more. Each of these interactions entails the transmission of data. Currently, the security of data transfer is ensured by the SSL protocol. However, possible vulnerabilities in data transfer include the following.

Data leakage: The SSL network security protocol assures data confidentiality and integrity during transmission. In addition, technologies such as data leakage prevention and advanced big data analytics can enhance the identification and prevention of data leaks, effectively fortifying the security of data transfer. Nevertheless, data leaks may manifest through various other means, including (Wang et al., 2023).

- Eavesdropping
- Man in the middle
- Packet sniffing
- Data interception

Intrusion: These attacks, often recognized as deliberate and unauthorized efforts, aim to compromise the security of computer systems, networks, or other information technology infrastructures. The primary goal is typically unauthorized access, data theft or manipulation, disruption of system operations, and jeopardizing the integrity, confidentiality, or availability of a system or network (Ahmad et al., 2019), (Molina Zarca et al., 2020). Most conventional intrusion attacks are capable of execution in an immersion environment. As a case in point, multiple attacks have targeted popular VR social learning systems (Valluripally et al., 2020). Examples include DDoS and associated attacks such as syn flood, eavesdropping, packet flooding, packet sniffing, packet tampering, and ping sweeping.

4.4. Data processing

AI and various Machine Learning (ML) techniques are emerging as fundamental technologies in the evolving metaverse. Stored data in the metaverse is utilized for training different models and enhancing the system. However, using ML techniques demands a high storage and processing capacity, which may not be feasible in many applications. Addressing this challenge involves offloading lightweight computations to terminals, and it also leads to the adoption of federated learning (Xu et al., 2023) and distributed learning, ensuring the privacy of users and IoT devices.

Attacking deep learning models: Deep Learning models in the metaverse are often large and complex, treated as black boxes, meaning we don't have a clear understanding of how predictions are extracted. Consequently, when these black-box models are deployed in a distributed manner on edge devices, attackers may potentially discover and exploit hidden issues. For instance, attackers could deceive the model into making incorrect predictions or disclose sensitive user information (Xiong et al., 2020).

Attacking federated learning models: In federated learning methods, there are also threats, including (Zhang, Zhu, et al., 2021).

- Data poisoning: In the metaverse, users may train ML models using their devices like smartphones or Head-Mounted Displays (HMDs). The inherently decentralized nature of federated learning leads to an unavoidable expansion of potential attack vectors. Malicious users might deliberately train the model on incorrectly labeled or manipulated data to deceive the global model.
- Attacks on edge nodes: Identifying the influential edge devices participating in federated learning enables attackers to launch an assault. They can compromise and take control of these devices, allowing them to manipulate the global model.
- Inference attacks: Federated learning training usually entails updating a global model using various local models on users' devices. With each update, adversarial participants can store model parameters in their local models, potentially transferring substantial information through these updates. This could include unintended features about participants' training data. As a result, attackers may deduce a significant amount of private information or even recover the original training samples of users through these model updates.
- Generative Adversarial Network (GAN) attack: This attack is among the most prevalent, generating deceptive data through the iterative merging of two neural networks. For instance, a deceptive participant posing as part of a project may try to extract weights from global model parameters and then regenerate sensitive information using a GAN attack (Li, Zhang, et al., 2023).

Malware: Embedded malware involves hackers utilizing the covert evil model to inject deceptive data into the training process (Wang et al., 2021). If developers don't have a solid grasp of algorithmic principles, they may not uncover the virus based on subtle differences in experimental results. Currently, unless the contaminated layer is frozen, the virus can be directly eliminated by altering parameter values and retraining the model. However, most developers may not modify parameters before training unless they intend to create other applications. Therefore, innovative approaches are needed to tackle these security threats.

4.5. Data exchange/transaction

Frequent data exchanges lead to the creation of intricate and intertwining data flow paths. The data lacks a singular directional flow and is not confined to a single system. Given that data can be copied and shared seamlessly, tracking them becomes a more formidable challenge. In the metaverse, data acts as input for DTs, and the resulting output feedback from these DTs initiates another data stream towards the user. Another set of data pertains to digital assets and their transactions within the blockchain. Threats within this realm include:

DTs data leakage: As DTs are typically based on real-world data, they often contain sensitive information such as personal health data, physiological characteristics, and organizational production data. For seamless interaction with these DTs in the metaverse, interactive terminals must gather data from them to simulate realistic scenarios. However, this process carries the risk of information leakage. If such information falls into the wrong hands, it could lead to significant privacy concerns (Dietz et al., 2019). Moreover, considering the virtual-real mapping established by DTs, the leakage of virtual data implies a leakage of real-world data, posing potential threats to life and property in the real world (Gehrmann & Gunnarsson, 2020).

DTs unauthorized tampering: Unauthorized access to DTs can enable attackers to manipulate or control their physical counterparts. This can lead to unauthorized actions or even harm the physical object or environment. Another crucial point is that DTs are often part of a larger IoT ecosystem and vulnerability in one part of the ecosystem can lead to security threats in DTs and vice versa (Han et al., 2023).

Information theft from DTs: DTs, particularly their humanoid forms known as avatars, function as representations of physical individuals in the metaverse. Numerous attacks that were previously applicable to human agents in Web 2.0, such as identity theft or impersonation, can now transpire in the metaverse and avatars. Furthermore, activities like tracking or espionage on avatars, exemplified by "man in the room" attacks, represent additional threats in the metaverse (Giaretta, 2025). Given that the behavioral patterns, preferences, habits, and activities of avatars in the metaverse can mirror those of their physical counterparts, attackers may collect digital footprints of avatars. Exploiting the related similarities with real users, they can facilitate accurate user profiling and potentially engage in illicit activities.

Man in the room and VR worm: The metaverse environment comprises various virtual rooms, some of which can be customized or locked to safeguard user privacy. The notion of man in the room involves an individual attempting illicit entry into these private spaces. In doing so, intruders may exploit existing vulnerabilities, discreetly enter rooms, and record all interactions of avatars within (Vondrek et al., 2022). Unauthorized access to rooms by violation of access controls can be another goal for attackers. Sometimes, man in the room attack occurs in the metaverse to spread a worm. A VR worm infects real users hosting VR, subsequently spreading among users who share the same virtual room.

Digital assets vulnerabilities: From the metaverse perspective, digital assets are divided into two categories (Lam et al., 2022).

- Native assets: These are assets born within the metaverse, gaining value exclusively in that realm without real-world equivalents.
- Blockchain-mapped assets: These assets are essential both in the metaverse and the real world, holding meaning within the blockchain (such as NFTs).

Beyond this classification, an alternative perspective (Pooyandeh et al., 2022) divides digital assets into four categories: art, gaming, collection, and utility. However, trading these assets on the blockchain may introduce privacy challenges for users (Galal and.Youssef, 2022). Conversely, attacks on the underlying smart contracts of the metaverse infrastructure could affect the ownership of these assets (Das et al., 2022). In some cases, it may even be challenging to trace the real owner of the asset (Li, Yao, et al., 2023).

Attacking smart contracts: The code employed in smart contracts for distributed applications is frequently untested. Prominent platforms like Ethereum and EOS serve as examples of cases where vulnerabilities in smart contracts have incurred significant losses. The 2016 DAO incident, arising from a smart contract vulnerability that led to the theft of millions of Ethers, severely damaged the Ethereum community (Chen et al., 2022).

4.6. Data destruction

The emergence of Web 2.0 and subsequent scandals like Cambridge Analytica led to the enactment of data protection laws initiated in 1998, which were later reinforced by the General Data Protection Regulation (GDPR) in 2018. This implies that the evolution towards a more comprehensive regulatory framework spanned approximately two decades. If this passive approach persists, the likelihood is that Web 3.0 will replicate not only similar but potentially more profound and

perilous mistakes observed in Web 2.0.

Metaverse governance by a handful of organizations: The concern is that, like the current internet, the metaverse will likely be controlled by a handful of powerful companies (Goldberg & Schär, 2023). Therefore, data must be ethically collected, adhering to agreements, and deleted according to protocols to prevent potential misuse. The concept of data destruction involves the thorough elimination of stored data using various technical tools to ensure irretrievability, especially when safeguarding critical information. Metaverse platforms should adopt a prudent approach in overseeing affiliated companies, providing the responsible utilization and sharing of data. Simultaneously, they must adhere to data owner licenses to properly manage the obliteration of data (Cheng, 2023).

Blockchain immutability: Given that the data infrastructure in the metaverse relies on blockchain technology, this aspect is less of a concern. Data recorded in the blockchain is inherently secure, as it cannot be modified or deleted (Afif Monrat et al., 2019). However, the immutability feature of the blockchain operates like a double-edged sword. If the recorded data becomes corrupted, compromises privacy, or violates regulations, it can present challenges, as removing such tainted data is not a straightforward task.

5. Security countermeasures

In the metaverse, data security becomes a vital prerequisite for growth and success. Security measures to counter the identified threats are divided into two main categories. The first category, known as cyber security behaviors, includes actions and habits that individuals and organizations in the metaverse adopt to protect themselves and their data, ensuring a safe and secure experience (Dunnett et al., 2023). This involves raising awareness about potential risks, learning how to use tools and techniques to mitigate these risks, and practicing strong security measures such as creating robust password and utilizing biometric and multi-factor authentication. Additionally, safeguarding personal information from phishing and virtual espionage, along with maintaining good security practices like regular software updates and the use of firewalls and antivirus software, is crucial (Al-Emran & Deveci, 2024). The second category focuses on securing the metaverse environment and implementing measures to counteract the identified threats. As studied in (Al-kfairy et al., 2024), the aim is to create a secure environment that enhances the trust of individuals and organizations, fostering a sense of safety and improving the quality of their experiences in the metaverse. Subsequently, this paper will elucidate security measures aimed at safeguarding its data by leveraging the underlying technologies of the metaverse. Fig. 5 provides a summary of security solutions.

5.1. Data acquisition security countermeasures

Authentication and integrity verification of input nodes: To establish data security, it is essential to authenticate and verify the integrity of the source as a first step (Qamar et al., 2023), (Kaur et al., 2023). For example (Tuan et al., 2022), provides a software-defined networking infrastructure with a programmable controller for decentralized authentication and access control for smart devices that connect to the metaverse. Some attempts to authenticate VR devices are based on private keys (Fotouhi et al., 2020), and some are based on private messages exchanged with devices (Sharma & Arya, 2022). In the following stage, the collection of precise and genuine data becomes imperative. This importance arises due to the considerable amount of data gathered in the metaverse originating from diverse and heterogeneous sources, posing challenges to data quality. The acquisition and verification of data integrity hold utmost significance, as data quality significantly influences nearly all tasks within the metaverse.

Management of edge nodes: Identifying the vulnerabilities of nodes, especially XR tools, is a key solution to this issue. Analyzing the APIs that connect these tools to the network, along with the data that can

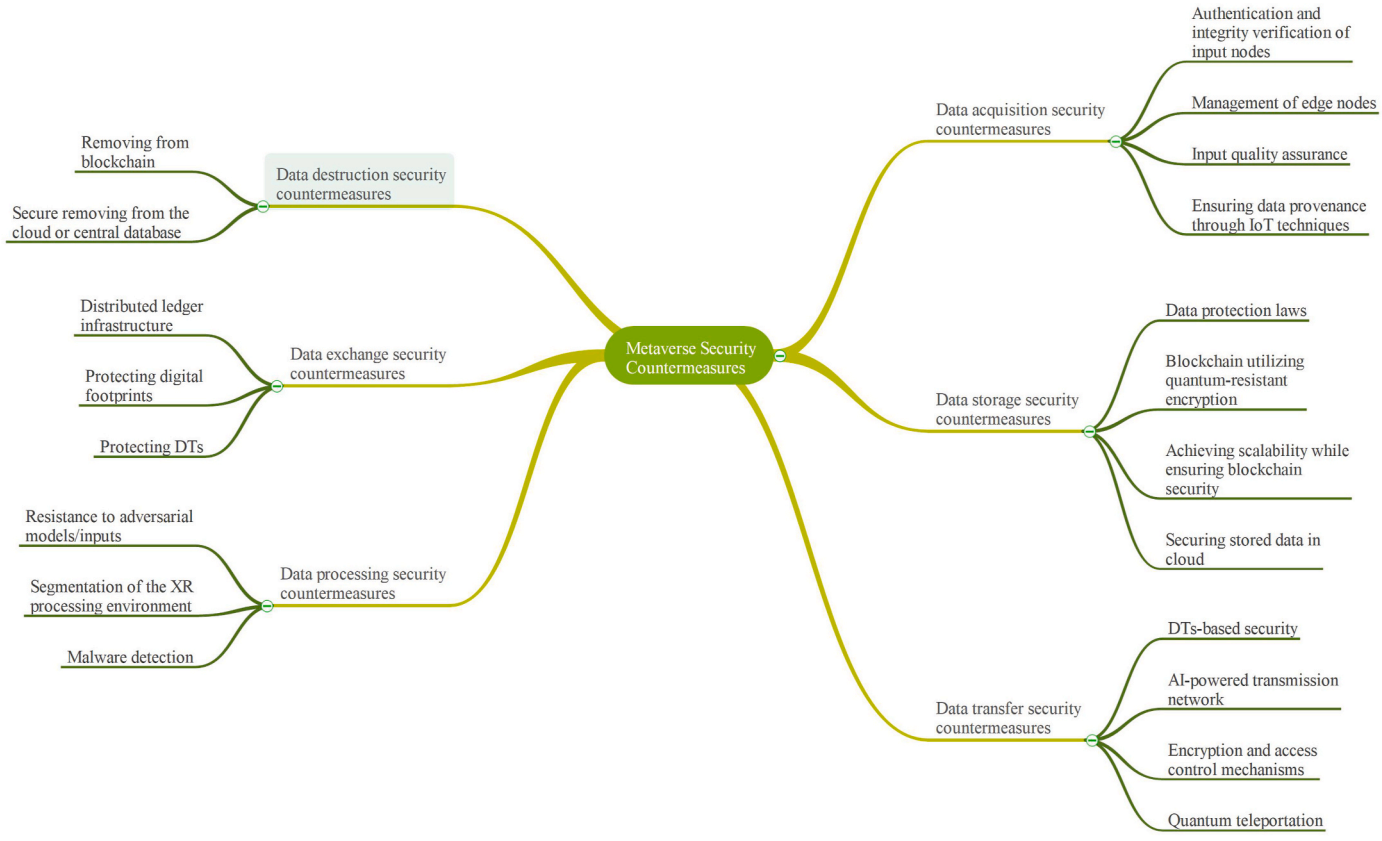


Fig. 5. Security threat countermeasures in the life cycle of the metaverse data.

be transferred between them and the virtual world, offers an effective approach to addressing these concerns (Guo et al., 2024). In addition, blockchain enables the automatic resource management of IoT devices and sensors through various smart contracts. Furthermore, it can motivate virtual space providers to facilitate the synchronization of virtual and physical realities by providing distributed incentives (Xu et al., 2023).

Input quality assurance: Ensuring the quality of input data is essential for the metaverse, as low-quality data from physical sensors and UGC by avatars can adversely impact both the quality of services and the user quality of experience. Effective quality control mechanisms play a crucial role in providing efficient metaverse services and maintaining the stability of the creator economy. One proposed solution involves leveraging AI for high-quality content generation and overcoming human limitations in production, coupled with the blockchain for content transfer and immutability (Lin et al., 2023). Additionally, considering the marketing of UGC and its economic value (Du et al., 2023), an approach to enhance its quality has been introduced. In addition, establishing specific criteria for web 3.0 content, similar to those proposed for web 2.0 (Mustoa & Dahanayake, 2021), is deemed necessary.

Ensuring data provenance through IoT techniques: Understanding the origin of data involves tracing the historical records of a specific piece of UGC. This process benefits evaluating data quality, tracking data sources, replicating the data generation process, and implementing a real-time audit trail to quickly identify responsible individuals. In the metaverse, information about the origin of UGC, such as its source, circulation, and intermediate processing details, is often stored in various data silos, such as distinct blockchain networks, posing a challenge for real-time monitoring and tracking. Lessons learned from existing work on data provenance in IoT can contribute valuable insights to the design of UGC provenance in the metaverse (Ardagna et al., 2020). Additionally, ongoing efforts to ensure data credibility and trustworthiness from edge nodes exemplify endeavors in this domain (Fan et al., 2021).

5.2. Data storage security countermeasures

Data protection laws: The Cybersecurity Law of the People's Republic of China, in effect since June 1, 2017, underscores the safeguarding of infrastructure and personal information. It introduces the principle of minimal necessary management, incorporates regulations on data breach notifications and individual deletion rights, and outlines rules for internal storage and assessment of personal information output. The GDPR in the European Union stands out globally, featuring both precise legal timelines and a well-organized legal framework. Implemented on May 25, 2018, GDPR enhances data rights and refines relevant mechanisms. In the United States, the California Consumer Privacy Act of 2018 (CCPA), enforced on January 1, 2020, adopts a profit-based data governance model. It sets consistent standards for protecting personal information, employing the "Regulations Protecting Public Data" model. This approach grants consumers comprehensive control over personal data while ensuring a balanced consideration of the interests of various industries (n.d. and Data Protection Laws of, 2023).

Blockchain resilient to quantum attacks using quantum-resistant encryption: As the number of quantum bits in quantum computers rises, the time needed to break classical encryption diminishes. Consequently, many researchers are concentrating on implementing quantum-resistant encryption to create a blockchain that can withstand quantum attacks. Existing quantum-resistant encryption algorithms primarily rely on the computational complexity of mathematical problems to resist quantum computer attacks. However, this computational complexity also introduces a substantial resource overhead on the network, posing challenges for adapting blockchain architecture (Zhang, Wang, et al., 2021).

Achieving scalability while ensuring blockchain security: Storing data on the blockchain eliminates the risk of SPoF in the metaverse (Seo & Park, 2024). Furthermore, blockchain-based access control can mitigate other risks, such as data leaks. However, scalability remains a significant limitation that hinders the widespread adoption of blockchain technology. Numerous strategies have been suggested to enhance blockchain scalability while preserving its security. Some of these approaches encompass reducing transaction sizes (Chan et al., 2021), transferring transactions to sidechains or off-chain (Jayabalan & Jeyanthi, 2022), boosting the block production rate through mechanisms like categorizing data types or other incentives (Sohrabi & Tari, 2020), parallelizing the block production and registration process by implementing node sharding (Manuskin et al., 2020).

Securing stored data in the cloud: Safeguarding data requires protective measures at every stage, including data storage and transmission. In data storage, it's crucial to encrypt information in a manner that prevents comprehension, even if hackers gain access to the data. Moreover, encryption with zero-knowledge proof allows storing users' encryption keys outside the cloud. This ensures that even if hackers breach the data server, they lack the decryption key, thwarting their attempts to read the data. Encryption renders conventional data recovery technologies ineffective, leading to the adoption of encrypted data recovery solutions such as linear search algorithms, security indexing, and public key keyword search algorithms (ThandaiahPrabu et al., 2022). To uphold the confidentiality and integrity of data, the data transfer should be encrypted using pertinent security protocols like PGP, SSL/TLS, IPSec, and others.

5.3. Data transfer security countermeasures

DTs-based security: Originally introduced as a foundational technology for the metaverse with their own unique security threats, DTs can play a crucial role in ensuring its security. Safeguarding DTs in the Cyber-Physical Systems (CPS) lifecycle involves multiple stages, including designing secure CPS, intrusion detection, identifying hardware and software misconfigurations, conducting security and privacy testing, system testing and training, safe deployment, and legal compliance. Integrating DTs into the metaverse can strengthen the interconnectedness between the virtual and real worlds (Eckhart et al., 2019). Since DTs exist solely in a virtual realm and often operate in an environment separate from ubiquitous systems, they can serve as experimental and educational platforms. Manufacturers can assess the security of their products or provide training on handling common cyber-attacks before deploying them. Analyzing the behavior of a system under imminent attack allows engineers to estimate potential vulnerabilities and threats. Additionally, another practical application of DTs is the assessment of strategies to mitigate damage to vulnerable devices, including the simulation of attack scenarios (B'ecue et al., 2018). Moreover, using DTs for creating intrusion detection systems is another valuable application in enhancing metaverse security (Eckhart & Ekelhart, 2018).

AI-powered transmission network: The use of AI to address attacks that may occur during data transmission to/from the metaverse has been explored in several studies (Awadallah et al., 2024). highlights that AI-based intrusion detection systems offer significant advantages over traditional methods, such as network intrusion detection systems or signature-based detection. These benefits include better adaptability to emerging threats and dynamic behaviors, the ability to detect zero-day attacks, effective pattern recognition for identifying threats in large datasets, real-time response, reduced false positives, and overall improved predictive accuracy. For instance (Ali et al., 2024), utilizes GANs to detect intrusions within the metaverse, generating synthetic data and training the system to identify abnormal data being exchanged between the real world and the metaverse. In addition, deep learning techniques are employed to distinguish between typical and anomalous user behaviors.

Encryption and access control mechanisms: The encryption of

data acts as a barrier against unauthorized access during the entire transmission process. Furthermore, integrating access control mechanisms, such as role-based and attribute-based controls, can strengthen the security of the transmission (Kang et al., 2023). Since the security of cryptographic approaches relies on the strength of encryption keys, the keys generated at each endpoint can be vulnerable to attacks. To mitigate this risk (Wang et al., 2024), proposes a distributed key generation method, employing an asynchronous approach tailored for the metaverse.

Quantum teleportation: Quantum teleportation, known for its ability to provide secure and reliable communication, is set to transform interactions within this immersive digital domain (Kumar et al., 2024). enables user-avatar interaction through a secure seven-qubit entangled state, while interactions between two avatars are facilitated via a two-qubit entangled state.

5.4. Data processing security countermeasures

Resistance to adversarial models/inputs: In the metaverse, the integration of AI, such as GANs, plays a pivotal role in generating dynamic gameplay scenarios and producing high-quality domain images. However, it introduces adversarial and poisoned security threats, creating instances that are challenging for human detection. In the existing literature, various strategies have been explored to enhance resilience against adversarial examples, including adversarial learning, adversarial representation learning, adversarial reinforcement learning, and adversarial transfer learning. These approaches involve incorporating adversarial samples into the training data to effectively counter adversarial threats and bolster the robustness of the metaverse (Soliman et al., 2023). Some examples include.

- To minimize the impact of attacks on edge nodes, each edge device's involvement in federated learning can be obscured using a randomization mechanism (with a minor performance trade-off). This mechanism incorporates random sub-sampling and perturbation. When the attacker is unaware of whether an edge device is engaged in training, the data from edge devices remains safeguarded (Rizk et al., 2022).
- A natural way to prevent privacy leakage involves adding artificial noise to the updated local model parameters before the global aggregation stage (Wei et al., 2020).
- To mitigate the impact of malicious network attacks, it is essential to improve the detection of malicious data. For this purpose, (Retiti Diop Emane et al., 2024) introduces an approach to recover original data by identifying and removing malicious nodes. Additionally, (Dai et al., 2019) enhances the model's generalization capability by incorporating disturbance samples into the training phase. (Li, Zhang, et al., 2023) employs a heuristic approach for identifying malicious data and relies on a firewall to prevent its dissemination.
- Discovering vulnerabilities in the ML pipeline, addressing security weaknesses, and improving the interpretability of modules are other examples of endeavors in this direction (Wang et al., 2017), (Li et al., 2021).
- To minimize the risk of model/data manipulation, training is outsourced to multiple third-party cloud platforms. During testing, a triple redundancy/multi-modular approach is employed for intrusion detection (Li et al., 2018).

Segmentation of the XR processing environment: In an AR/VR device, we encounter two distinct processing environments—Trusted Execution Environment (TEE) and Rich Execution Environment (REE). TEE, characterized by secure and reliable processing, memory, and storage capabilities, operates separately. On the other hand, the device's operating system and metaverse applications run in REE. To address security concerns, the device's operating system strategically partitions metaverse applications, limiting operations and safeguarding sensitive data within the TEE. The TEE implementation can streamline identity verification by eliminating the need for hardware tokens. Intel's Software Guard Extensions (SGX) is an example of software security extensions designed for the Intel architecture, leveraging the TEE to isolate program code and specific data in memory (Costan & Devadas, 2016).

Malware detection: The utilization of ML techniques to identify malware represents ongoing efforts in this field (Demontis et al., 2019). Nevertheless, adversaries can enhance their capabilities through poisoning attacks or evasion strategies. Strategies for improving adversarial learning models and converging their outcomes in detecting evasive techniques have been outlined in (Li, Cui, et al., 2023).

5.5. Data exchange security countermeasures

Distributed ledger infrastructure: Considering the characteristics of blockchain (transparency, immutability, irrevocability, decentralization, etc.), key aspects in managing and securing the metaverse data include terminal hosting, application systems, network boundaries, and so on; these aspects must be addressed appropriately. These key points can be preserved through effective management, control, and certification by blockchain storage and smart contracts to ensure the integrity, accuracy, and confidentiality of data (Li, Yao, et al., 2023). However, the blockchain will bring various security benefits to the metaverse (Xu et al., 2023). For instance, UGC crafted within the metaverse can be permanently recorded and stored in the blockchain as NFTs, establishing verifiable and unique ownership of UGC. This incentivizes users to actively contribute content, thereby maintaining the vibrancy of the metaverse. Digital content can be traded within the economic framework of the metaverse using digital currencies or fiat.

Protecting digital footprints: This solution, designed to evade virtual espionage and man in the room attack, involves diverse strategies such as creating distance in the virtual environment (n.d. Meta Establishes 4, 2022), hiding avatar (Vondrek et al., 2022), generating various avatar clones, and periodically altering the appearance of avatars (Ning et al., 2023). Other methods include making avatars invisible, confining them within personalized environments, and locking down those environments (Falchuk et al., 2018). While solutions like locked personalized environments and creating distance appear effective in avoiding VR worms, they still require testing.

Protecting DTs: As previously explained, DTs can serve as educational and experimental platforms to understand various attacks that might occur in the metaverse. However, ensuring the security of the input data transferred to DTs and preventing alterations or leaks involves tasks such as encryption, secure storage, authorized access control, and regular backups (Dietz et al., 2019).

5.6. Data destruction security countermeasures

Removing from the blockchain: Several alternative approaches have been proposed for deleting data recorded in the blockchain. Nevertheless, these methods are currently in the proposal stage and have not been implemented. The introduction of redactable blockchain technology is an endeavor undertaken for this purpose (Abd Ali et al., 2023). (Hou et al., 2021) has introduced a technique for purging detrimental data from the blockchain, necessitating a renewed consensus among the participating nodes. The incentive reward is bestowed upon the purging miner by the node responsible for transmitting the flawed transaction across the network.

Secure removal from the cloud or central database: Certain data related to the metaverse system or its users might be stored in the cloud or centralized databases. In these instances, it becomes crucial to adopt a suitable methodology, governed by the data owner's license, to effectively eliminate such data.

6. Discussion

By bridging the gap between theoretical cybersecurity frameworks and the distinct, evolving challenges of the metaverse, this study not only enriches academic discourse but also offers practical guidance for safeguarding next-generation digital ecosystems. Theoretically, by introducing a structured classification of security threats across distinct phases of the data lifecycle in the metaverse, this research offers a novel lens through which emerging cybersecurity challenges can be understood and analyzed. This lifecycle-based approach moves beyond traditional threat-centric models by integrating insights from underlying infrastructure technologies, thereby laying a robust conceptual foundation for future academic inquiry. Moreover, this interdisciplinary perspective enhances the theoretical discourse on data governance and security in the context of emerging digital ecosystems like the metaverse.

Practically, the findings of this study serve as a valuable guide for developers, platform designers, and policymakers by offering actionable insights into where and how security interventions can be most effective throughout the data lifecycle—from acquisition to destruction. The proposed security countermeasures can inform the design of robust security protocols, enhance regulatory compliance, and support more effective risk management in metaverse applications. Additionally, the research can be adapted as a practical tool for security training and awareness programs, helping stakeholders better anticipate, detect, and respond to evolving threats in immersive environments.

Although a variety of security threats and their corresponding countermeasures have been examined, the dynamic nature of data management within metaverse environments presents new, multifaceted challenges. These challenges require innovative, proactive strategies that not only safeguard user data but also ensure its integrity, confidentiality, and resilience in an ever-evolving virtual ecosystem.

The data volume is expected to be significantly larger than currently in the ideal metaverse scenario. This is primarily due to the anticipation of increased interactive modes and the proliferation of more terminals, accompanied by the introduction of new technologies that bring about numerous unforeseen applications and states, making their prediction quite challenging. For example, if the data generated during autonomous driving by innovative vehicles is integrated into the metaverse, or if all products sold in physical stores have a virtual presence, the data volume will be extremely substantial. Managing this diverse and voluminous data goes beyond conventional big data processing and introduces unique security challenges.

The metaverse operates as a data-centric environment, and overcoming challenges related to data processing is paramount. Initially, attention must be given to issues concerning data storage and access. This not only involves catering to a vast number of users accessing data from diverse locations but also managing numerous and varied access requests to the system, along with facilitating data exchange. The presence of various data types in stored data (2D and 3D) and applying contextual queries to the system can significantly impact the retrieval of information within the metaverse (Zhang et al., 2024). Additionally, the ability to respond promptly in data processing is of utmost importance, as prolonged delays are incompatible with the all-encompassing design of the metaverse (Chen et al., 2024).

In the current iteration of the internet, we are witnessing the profound implications of entrusting societal responsibilities to algorithms. Despite their exceptional performance, these algorithms grapple with several challenges, including biases impeding equitable outcomes, a lack of transparency, susceptibility to various attacks and manipulations, and the substantial computational and energy demands associated with complex AI models. These limitations not only hinder their cost-effectiveness but also impact their sustainability. Each issue represents a pressing scientific challenge that necessitates collaborative efforts across diverse scientific domains, encompassing AI, ML, security, ethics, and more, to comprehensively address and resolve (Pietro & Cresci, 2022).

As a practical and pertinent illustration of the complications stemming from algorithms and automation, security concerns related to the amalgamation of content and user identity verification can be considered. Even within contemporary social networking platforms, a considerable portion of our interactions involves non-organic or synthetic content, such as text generated by machines, and engagements with untrustworthy users, like fake personas operated by humans for trolling or entirely automated accounts such as bots. In the metaverse, these human-machine interactions are anticipated to become more frequent and, in certain activities, mandatory. These evolving capabilities bring about new and intensified security considerations.

The final and crucial aspect concerns the notion of darkverse. In the metaverse, the darkverse shares similarities with the dark web on the internet. Due to the quasi-physical presence of users, the darkverse has the potential to pose even greater hazards than the dark web. Moreover, the absence of clear indicators complicates their control (Chen et al., 2022). The role of community leaders and the adoption of innovative approaches, such as gamification, for user education and awareness (Sreerakuvandana et al., 2024), are key strategies that can complement technical and legal measures in mitigating potential harms in this environment.

7. Conclusion

The objective of this article was to examine the security threats in the metaverse from the standpoint of data security governance. The dual-edged sword nature of technology appears to persist in the metaverse. While advancements in technologies like AI, DT, and IoT contribute to the creation of the metaverse, they also expose this space to new attacks. The proposed solution in the article, addressing these attacks, once again relies on utilizing emerging technologies such as AI and blockchain. This establishes a cycle where technology is employed to resolve vulnerabilities, only to encounter new vulnerabilities in the process. The results of this study are expected to support all metaverse stakeholders—including users, AR/VR device manufacturers, developers, and service providers—by increasing awareness of these vulnerabilities and guiding them in implementing effective countermeasures. Furthermore, specialists and researchers in the foundational technologies should focus on finding solutions to address the weaknesses of these technologies or explore opportunities to combine them with other innovative technologies.

While this investigation aimed to analyze security incidents occurring throughout the metaverse data lifecycle, utilizing existing documents and examining implemented metaverse instances, it's crucial to note that the metaverse, as described with features like immersion, ubiquity, and absence of temporal and spatial limitations, has not been fully realized. Moreover, the accessibility of metaverse connectivity tools, such as HMDs and other devices, remains challenging due to their high cost, limiting widespread adoption. This industry's slower adoption has resulted in many potential risks not evolving into practical threats. Consequently, the study of threats is confined to theoretical scenarios. Additionally, the maturation of underlying metaverse technologies is ongoing, potentially introducing new security vulnerabilities and solutions. Nevertheless, a practical investigation of the security threats associated with the six-stage data lifecycle, conducted in a virtual world similar to the metaverse, remains a key research objective for future work. In parallel, we will focus on studying security threats related to access control and privacy, specifically from the perspective of metaverse users. Additionally, exploring the industrial metaverse to identify potential vulnerabilities will be a priority in the near future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

No data was used for the research described in the article.

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